Chapter 3

Basic Definitions

Set theory is the foundation of all branches of mathematics. It is therefore important to give it a rigorous treatment in our study of set theory. This means that every concept used in the theory should be defined and any claim made in the theory should be justified by a formal proof. Since the concept of *set* is the main subject of set theory, it is a concept we have to define. So, our first question is:

What is a set?

A common answer to the above question is: "a set is a collection of objects." Unfortunately, this answer is far from satisfactory, for in attempting to define the concept of *set*, it uses the concept of *collection* whose definition has not been given. What the answer has done is essentially changing the question from "What is a set?" to "What is a collection?" So, our question becomes:

What is a collection?

A moment of reflection reveals that to define the concept *collection*, we have to come up with another concept. Whatever this concept is, we will immediately be confronted with the same problem, that is, to give that concept a definition.

Therefore, the problem we are facing is that each time we come up with a new concept to define a given concept, we immediately have to find a definition for that new concept. Obviously, this can be repeated indefinitely. So, even before we can start our study of set theory, we are already stuck with a non-terminating chain of questions.

The moral of the story is that although it would be ideal if we could define every concept in our study of set theory, it turns out to be impossible to achieve that. We simply cannot develop something out of nothing. We must begin our study with some concepts which need no definition. These concepts are called ***basic concepts***. All other concepts are then defined based on the basic concepts or concepts which have been previously defined. Clearly, each basic concept should be primitive enough so that one will have little doubt as to what its meaning is. Moreover, the number of basic concepts used in the study should be kept to a minimum.

In set theory, the concepts of ***set***, ***element*** and ***element of*** are the basic concepts.

3.1 Basic Notations

Owing to the aforementioned reasons, the notation, $x \in A$, is a valid notation that needs no definition. The variable x represents an element, the symbol $\in$ represents the concept 'element of' and the variable

A represents a set. Elements are usually represented by lower-case letters such as x, y, z while sets are usually represented by upper-case letters such as A, B, C. So $x \in A$ denotes x is an element of A, and can be used as a predicate in FOL-wffs. The notation $x \notin A$ denotes $\sim x \in A$. Moreover, if $(x_1 \in A)$ and $(x_2 \in A)$ and . . . and $(x_n \in A)$, we shall use the abbreviation $x_1, x_2, \cdots, x_n \in A$. In the same spirit, $x_1, x_2, \cdots, x_n \notin A$ is an abbreviation of $(x_1 \notin A)$ and $(x_2 \notin A)$ and . . . and $(x_n \notin A)$.

We sometimes say that x *belongs to* A, or x is a *member of* A, or A *contains* x, instead of saying x is an element of A.

Example 3.1.1 The following are some examples of sets:

(i) The set of English vowers a, e, i, o, u.

(ii) The set consisting of the following elements:

the set consisting of the integers 1 and 2;
the set consisting of the letters a, b and c;
the Greek character α, and
the letters x, y z.

(iii) The set of prime numbers smaller than 50.

(iv) The set of provinces in Canada.

(v) The set of universities in North America.

(vi) The set of rational numbers.

(vii) The set of points at a distance r from a fixed point o in the plane.

Note that (ii) shows that an element of a set can be a set itself.

Example 3.1.2

Let A denote the set in Example 3.1.1(i). Then $a \in A$, $b \notin A$ and $e, i, u \in A$.
Let B denote the set in Example 3.1.1(iii). Then $7 \in B$, $12 \notin B$ and $2, 13, 29, 47 \in B$.
Let $\mathbf{Q}$ denote the set in Example 3.1.1(vi). Then $7, \frac{2}{3}, -1.34 \in \mathbf{Q}$, and $\pi, \sqrt{2} \notin \mathbf{Q}$.

For conciseness and preciseness, we need a formal method to specify (or define) sets. Two methods are commonly used. The first method is the ***brace notation***, which enumerates the elements of the set within a pair of braces (i.e. { }), with commas separating the elements. The following are some examples.

Example 3.1.3 Using the brace notation, the first four sets in Example 3.1.1 can be specified as follows.

(i) $\{a, e, i, o, u\}$.

(ii) $\{\{1, 2\}, \{a, b, c\}, \alpha, x, y, z\}$.

(iii) $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}$

(iv) $\{Ontario,\ Quebec,\ British{-}Columbia,\ Alberta,\ Saskatchewan,\ Manitoba,\ New\ Brunswick,$
$Nova\ Scotia,\ Prince\ Edward\ Island,\ Newfoundland\}$

Unfortunately, for the set in Example 3.1.1(v), it is not as easy to write out the set specification using the brace notation, for there are a few thousand universities in North America. When we get down to the last two sets, the situation gets even worse as each of them contains infinitely many elements. We just cannot enumerate all of their elements.

Therefore, when the number of elements in a set is very large or the set has infinitely many elements, the brace notation becomes inconvenient or useless. A more powerful set specification method is needed. The second method that is commonly used is called the ***set-builder notation.*** It will be discussed in Section 3.3.

Definition: A set having exactly one element is called a ***singleton***. A set which has exactly two elements is called an ***unordered pairs***.

The simplest form of singleton and unordered pair are $\{a\}$ and $\{a, b\}$, respectively. Other examples of singletons are $\{\{a\}\}$ and $\{\{\{w\}\}\}$, and of unordered pairs are $\{\{a\}, \{a, b\}\}$ and $\{a, \{a\}\}$.

3.2 Set Equality and Set Inclusion

Let A be a set. Given any object x, either x is an element of A, or x is not an element of A, but not both. So, a set is uniquely determined by its elements. As a result, two sets are identical if and only if they have identical elements. This is called the Principle of Extension, which is formally defined as follows.

Definition: (***Principle of Extension***) Two sets A and B are ***equal***, denoted by $A = B$, iff $(\forall x)(x \in A \Leftrightarrow x \in B)$.

The negation of $A = B$ is denoted by $\sim (A = B)$ or $A \neq B$.

In mathematics, it is a common practice to omit the universal quantifier from wffs of the form $(\forall x)\alpha(x)$ provided that no confusion could occur in doing so. Therefore, the wff $(\forall x)(x \in A \Leftrightarrow x \in B)$ in the above definition is commonly abbreviated as $(x \in A \Leftrightarrow x \in B)$. We shall follow this common practice whenever is convenient in the remaining part of this notes.

We shall use the Principle of Extension to prove the following basic properties of set equality.

Lemma 3.2.1: Let A, B, C be sets.

(i) $A = A$

(ii) If $A = B$, then $B = A$.

(iii) If $A = B$ and $B = C$, then $A = C$.

[Remarks:

Before presenting the proof, we want to point out that all the formal proof methods developed in first-order logic remain valid in Set Theory. Moreover, we shall outline a number of common practices which we will follow in presenting proofs in the rest of these notes.

First, when no ambiguity could occur, short English sentences or informal but precise expressions are commonly used in place of FOL-wffs.

Second, let $\alpha_1, \alpha_2, \cdots, \alpha_n$ be a chain of reasoning such that each α_i is a logical consequence of its immediate predecessor α_{i-1}. In presenting the chain of reasoning, we customary use the implication

operator $\Rightarrow$ to chain the α_i's together, resulting in:

$$\begin{array}{ll} & \alpha_1 \\ \Rightarrow & \alpha_2 \qquad \gamma_2 \\ & \vdots \\ \Rightarrow & \alpha_n \qquad \gamma_n \end{array}$$

Each γ_i is the justification for the claim that α_i is a logical consequence of its immediate predecessor α_{i-1}. Note that the α_i's need not be listed in a strictly vertical order as shown above.

When the reverse chain of reasoning, $\alpha_n, \alpha_{n-1}, \cdots, \alpha_1$, is also valid, we customary use the biconditional operator $\Leftrightarrow$ to chain the α_i's together, resulting in:

$$\begin{array}{ll} & \alpha_1 \\ \Leftrightarrow & \alpha_2 \qquad \gamma_2 \\ & \vdots \\ \Leftrightarrow & \alpha_n \qquad \gamma_n \end{array}$$

Each γ_i is the justification for the claim that α_i is a logical consequence of its immediate predecessor α_{i-1} and vice versa. Again, the α_i's need not be listed in a strictly vertical order.

Third, in using the bidirectional proof method to prove a theorem of the form $\vdash (\alpha \Leftrightarrow \beta)$, the symbol $\Rightarrow$) is used to mark the beginning of the proof for $\vdash (\alpha \Rightarrow \beta)$, while the symbol $\Leftarrow$) is used to mark the beginning of the proof for $\vdash (\beta \Rightarrow \alpha)$.

Fourth, owing to the Soundness and Completeness Theorem of first-order logic, whenever a theorem of the form $\vdash (\alpha \Leftrightarrow \beta)$ is proven, the validity of $(\alpha \Leftrightarrow \beta)$ is also established. This means that we have established the equivalence $\alpha \equiv \beta$. As a result, we can feel free to substitute α for β or β for α in subsequent proofs. For a similar reason, once we have proven a theorem of the form $\vdash (\alpha \Rightarrow \beta)$, we may feel free to substitute β for α (but not α for β) in subsequent proofs.

Fifth, we shall use "and" and $\wedge$ ("or" and $\vee$, "iff" and $\Leftrightarrow$, respectively) interchangably.

The following extension of Inference rule I3 can be useful in shortening proofs.

$\mathbf{I3}_\wedge$: $\alpha \wedge \beta, \alpha \Rightarrow \gamma \vdash \gamma \wedge \beta$

$\mathbf{I3}_\vee$: $\alpha \vee \beta, \alpha \Rightarrow \gamma \vdash \gamma \vee \beta$

For their correctness, see Section 1.9, Exercises 3m and 3n.]

Proof of Lemma 3.2.1:

(i) We are to prove that $(\forall x)(x \in A \Leftrightarrow x \in A)$.

$(x \in A \vee x \notin A)$ (Axiom)

$\Rightarrow (x \notin A \vee x \in A)$ (E10)

$\Rightarrow (x \in A \Rightarrow x \in A)$ (E18)

$\Rightarrow (x \in A \Rightarrow x \in A) \wedge (x \in A \Rightarrow x \in A)$ (E3)

$\Rightarrow (x \in A \Leftrightarrow x \in A)$ (E20)

$\Rightarrow (\forall x)(x \in A \Leftrightarrow x \in A)$ (Gen)

Hence, $A = A$. (Principle of Extension)

(ii) (Direct proof) Suppose $A = B$.

Then $(\forall x)(x \in A \Leftrightarrow x \in B)$ (Principle of Extension)

$\Rightarrow (\forall x)(x \in B \Leftrightarrow x \in A)$ (Theorem 1.7.2(ii))

Hence, $B = A$. (principle of Extension)

(iii) (Direct proof) Suppose $A = B$ and $B = C$.

Then $(\forall x)(x \in A \Leftrightarrow x \in B)$ and $(\forall x)(x \in B \Leftrightarrow x \in C)$. (Principle of Extension)

$\Rightarrow (x \in A \Leftrightarrow x \in B)$ and $(x \in B \Leftrightarrow x \in C)$ (UI)

$\Rightarrow (x \in A \Leftrightarrow x \in B) \wedge (x \in B \Leftrightarrow x \in C)$ (I6)

$\Rightarrow (x \in A \Leftrightarrow x \in C)$ (Theorem 1.7.2(iv))

$\Rightarrow (\forall x)(x \in A \Leftrightarrow x \in C)$ (Gen)

Hence, $A = C$ (Principle of Extension). ■

Suppose $A_1 = A_2$ and $A_2 = A_3$ and . . . and $A_{n-1} = A_n$. Owing to Part (iii) of the above theorem, it makes sense to write $A_1 = A_2 = \cdots = A_n$ as an abbreviation of $A_i = A_j$, for $1 \leq i < j \leq n$.

The following lemma is useful in proving that two sets are not equal.

Lemma 3.2.2: $A \neq B$ *iff* $(\exists x)(x \in A \wedge x \notin B) \vee (\exists x)(x \notin A \wedge x \in B)$.

Proof: $A \neq B \Leftrightarrow \sim (\forall x)(x \in A \Leftrightarrow x \in B)$ (Principle of Extension)

$\Leftrightarrow \sim (\forall x)((x \in A \Rightarrow x \in B) \wedge (x \in B \Rightarrow x \in A))$ (E20)

$\Leftrightarrow (\exists x) \sim ((x \in A \Rightarrow x \in B) \wedge (x \in B \Rightarrow x \in A))$ (FE7)

$\Leftrightarrow (\exists x)(\sim (x \in A \Rightarrow x \in B) \vee \sim (x \in B \Rightarrow x \in A))$ (E16)

$\Leftrightarrow (\exists x)(\sim (\sim x \in A \vee x \in B) \vee \sim (\sim x \in B \vee x \in A))$ (E18)

$\Leftrightarrow (\exists x)((\sim\sim x \in A \wedge \sim x \in B) \vee (\sim\sim x \in B \wedge \sim x \in A))$ (E17)

$\Leftrightarrow (\exists x)((x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A))$ (E15)

$\Leftrightarrow (\exists x)((x \in A \wedge x \notin B) \vee (x \notin A \wedge x \in B))$ (E9)

$\Leftrightarrow (\exists x)(x \in A \wedge x \notin B) \vee (\exists x)(x \notin A \wedge x \in B)$ (FE10) ■

In other words, two sets A and B are unequal iff either set A contains something set B does not have, or set B contains something set A does not have.

Corollary 3.2.2.1: *Let A and B be sets.* $(\exists x)(x \in A \wedge x \notin B) \Rightarrow A \neq B$.

Proof: (Direct proof) $(\exists x)(x \in A \wedge x \notin B) \Rightarrow (\exists x)(x \in A \wedge x \notin B) \vee (\exists x)(x \notin A \wedge x \in B)$ (I1)

$\Rightarrow A \neq B$ (Lemma 3.2.2) ■

Example 3.2.4

(i) Prove that $\{a, b, c\} = \{b, c, a\}$.

$x \in \{a, b, c\} \Leftrightarrow (x = a)$ *or* $(x = b)$ *or* $(x = c) \Leftrightarrow x \in \{b, c, a\}$

Therefore, $\{a, b, c\} = \{b, c, a\}$ (Principle of Extension).

This example shows that when using the brace notation, the order in which the elements are listed is unimportant.

(ii) Prove that $\{a, b, c\} = \{a, a, b, c\}$.

The proof is similar to that for Part (i).

This example shows that when using the brace notation to denote a set, it makes no sense to duplicate any element in the set. In other words, every element in a set is distinct.

(iii) Prove that $\{\{a\}\} \neq \{a\}$.

Since $\{a\} \in \{\{a\}\}$ and $\{a\} \notin \{a\}$, we have

$(\{a\} \in \{\{a\}\}) \wedge (\{a\} \notin \{a\})$ (I6)

$\Rightarrow (\exists x)(x \in \{\{a\}\} \wedge x \notin \{a\})$ (EQ)

$\Rightarrow (\exists x)(x \in \{\{a\}\} \wedge x \notin \{a\}) \vee (\exists x)(x \in \{a\} \wedge x \notin \{\{a\}\})$ (I1)

$\Rightarrow \{\{a\}\} \neq \{a\}$ (Lemma 3.2.2) ■

Remark: By the Principle of Extension, $A = B$

$\Rightarrow (\forall x)(x \in A \Leftrightarrow x \in B)$

$\Rightarrow (\forall x)((x \in A \Rightarrow x \in B) \wedge (x \in B \Rightarrow x \in A))$ (E20)

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in B) \wedge (\forall x)(x \in B \Rightarrow x \in A)$ (FE9)

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in B)$ (I2)

This gives rise to the right substitutivity of $\in$. By including the three axioms for equality and the following right substitutivity of $\in$ in our discussion,

$$\vdash \quad (\forall x)(\forall y)(\forall A)((x = y) \Rightarrow (x \in A \Rightarrow y \in A)),$$

a set theory as a first-order theory with equality can be developed. This confirms our claim that all the formal proof methods developed in first-order logic remain valid in set theory.

Definition: Let A, B be two sets. A is a ***subset*** of B or B ***includes*** A, denoted by $A \subseteq B$, iff $(\forall x)(x \in A \Rightarrow x \in B)$. A is a ***proper subset*** of B, denoted by $A \subset B$, if $A \subseteq B$ and $A \neq B$.

The notation $A \nsubseteq B$ denotes $\sim(A \subseteq B)$ and the notation $A \not\subset B$ denotes $\sim(A \subset B)$. $A \subseteq B \subseteq C$ is an abbreviation of $(A \subseteq B) \wedge (B \subseteq C)$.

Example 3.2.5

(i) $\{a, u\} \subseteq \{a, e, i, o, u\}$

(ii) $\{a, b, c\} \subseteq \{a, b, c\}$

(iii) Let $\mathbf{R}$ be the set of all real numbers;

$\mathbf{Q}$ be the set of all rational numbers;

$\mathbf{Z}$ be the set of all integers;

$\mathbf{N}$ be the set of all positive integers;

$\mathbf{I}^-$ be the set of all negative integers;

$\mathbf{E}$ be the set of all even integers.

Since every positive integer is an integer; every integer is a rational number, and every rational number is a real number, we have $\mathbf{N} \subseteq \mathbf{Z}$, $\mathbf{Z} \subseteq \mathbf{Q}$ and $\mathbf{Q} \subseteq \mathbf{R}$.

Similarly, as every negative integer and every even integer is an integer, we have $\mathbf{I}^- \subseteq \mathbf{Z}$ and $\mathbf{E} \subseteq \mathbf{Z}$.

However, as *not* every even integer is a positive integer and not every even integer is a negative integer, we have $\mathbf{E} \nsubseteq \mathbf{N}$ and $\mathbf{E} \nsubseteq \mathbf{I}^-$.

Since *no* positive integer is a negative integer and vice versa, therefore $\mathbf{N} \nsubseteq \mathbf{I}^-$ and $\mathbf{I}^- \nsubseteq \mathbf{N}$. ■

Lemma 3.2.3: Let A, B, C be sets.

(i) $A \subseteq A$

(ii) If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

(iii) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Proof:

(i) We are to prove that: $(\forall x)(x \in A \Rightarrow x \in A)$.

$(x \in A \vee x \notin A)$ (Axiom)

$\Rightarrow (x \notin A \vee x \in A)$ (E10)

$\Rightarrow (x \in A \Rightarrow x \in A)$ (E18)

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in A)$ (Gen)

Hence, $A \subseteq A$. (Definition of $\subseteq$)

(ii) (Direct proof) Suppose $A \subseteq B$ and $B \subseteq A$.

Then, $(\forall x)(x \in A \Rightarrow x \in B)$ and $(\forall x)(x \in B \Rightarrow x \in A)$ (Definition of $\subseteq$)

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in B) \wedge (\forall x)(x \in B \Rightarrow x \in A)$ (I6)

$\Rightarrow (\forall x)((x \in A \Rightarrow x \in B) \wedge (x \in B \Rightarrow x \in A))$ (FE9)

$\Rightarrow (\forall x)(x \in A \Leftrightarrow x \in B)$ (E20)

$\Rightarrow A = B$. (Principle of Extension)

(iii) (Direct proof) Suppose $A \subseteq B$ and $B \subseteq C$.

Then, $(\forall x)(x \in A \Rightarrow x \in B)$ and $(\forall x)(x \in B \Rightarrow x \in C)$ (Definition of $\subseteq$)

$\Rightarrow (x \in A \Rightarrow x \in B)$ and $(x \in B \Rightarrow x \in C)$ (UI)

$\Rightarrow (x \in A \Rightarrow x \in C)$ (I5)

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in C)$ (Gen)

$\Rightarrow A \subseteq C$. (Definition of $\subseteq$) ■

Suppose $A_1 \subseteq A_2$ and $A_2 \subseteq A_3$ and . . . and $A_{n-1} \subseteq A_n$. Owing to Part (iii) of the above Lemma, it makes sense to write $A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n$ as an indication that $A_i \subseteq A_j$, for $1 \leq i < j \leq n$. Moreover, owing to Exercise 3.8.1, we may replace some or all of the $\subseteq$'s in the notation by the $\subset$ symbol.

Lemma 3.2.4: Let A, B be sets. If $A = B$, then $(A \subseteq B) \wedge (B \subseteq A)$.

Proof: $A = B \Rightarrow (\forall x)(x \in A \Leftrightarrow x \in B)$ (Principle of Extension)

$\Rightarrow (\forall x)((x \in A \Rightarrow x \in B) \wedge (x \in B \Rightarrow x \in A))$ (E20)

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in B) \wedge (\forall x)(x \in B \Rightarrow x \in A)$ (FE9)

$\Rightarrow (A \subseteq B) \wedge (B \subseteq A)$ (Definition of $\subseteq$) ■

Lemma 3.2.5: $A \nsubseteq B \Leftrightarrow (\exists x)(x \in A \wedge x \notin B)$.

Proof: $A \nsubseteq B \Leftrightarrow \sim (\forall x)(x \in A \Rightarrow x \in B)$ (Definition of $\subseteq$)

$\Leftrightarrow (\exists x) \sim (x \in A \Rightarrow x \in B)$ (FE7)

$\Leftrightarrow (\exists x) \sim (x \notin A \vee x \in B)$ (E18)

$\Leftrightarrow (\exists x)(\sim x \notin A \wedge x \notin B)$ (E17)

$\Leftrightarrow (\exists x)(x \in A \wedge x \notin B)$ (E15) ■

Lemma 3.2.6: If $A \subset B$, then $(\exists x)(x \in B \wedge x \notin A)$.

Proof: (Direct proof)

$A \subset B \Rightarrow (A \subseteq B) \wedge A \neq B$ (Definition of $\subset$)

$\Rightarrow (A \subseteq B)\ and\ (A \neq B)$ (I2,E9)

Now,

$A \neq B \Rightarrow (\exists x)(x \in A \wedge x \notin B) \vee (\exists x)(x \in B \wedge x \notin A)$ (Lemma 3.2.2)

$\Rightarrow \sim\sim(\exists x)(x \in A \wedge x \notin B) \vee (\exists x)(x \in B \wedge x \notin A)$ (E15)

$\Rightarrow \sim (\exists x)(x \in A \wedge x \notin B) \Rightarrow (\exists x)(x \in B \wedge x \notin A)$ (E18) . . . . . (I)

and

$A \subseteq B \Rightarrow \sim A \nsubseteq B$ (E15)

$\Rightarrow \sim (\exists x)(x \in A \wedge x \notin B)$ (Lemma 3.2.5) . . . . . (II)

We thus have $(\exists x)(x \in B \wedge x \notin A)$ ((I),(II), I3) ■

3.3 Set Specifications

In Section 3.1, we experienced difficulty in specifying sets using the brace notation when a given set has a huge number of elements. In this section, we shall study a more powerful method of specifying sets, called the ***set-builder notation.*** In this method, a set is specified by a property possessed by all its elements and only its elements. This property is called a ***defining property***. Defining properties are expressed as formulas defined as follows.

Definition: A ***formula*** is a FOL well-formed formula containing a free variable and may contain expressions in the argument lists of its predicates.

Example 3.3.6 For each of the sets in Example 3.1.1, identify a defining property for the set and express it as a formula.

(i) A defining property is: "an object x is an element of the set iff it is one of the letters a, e, i, o, u." This property can be expressed as $(x = a) \vee (x = e) \vee (x = i) \vee (x = o) \vee (x = u)$, where x is the only free variable in the formula. The letters a, e, i, o, u are constants.

(ii) A defining property is: "an object u is an element of the set iff it is one of the following objects $\{1, 2\}, \{a, b, c\}, \alpha, x, y, z$." Since the number of elements in the set is small, we may use a formula similar to that in part (i) to express the above defining property. The formula is: $(u = \{1, 2\}) \vee (u = \{a, b, c\}) \vee (u = \alpha) \vee (u = x) \vee (u = y) \vee (u = z)$, where u is the free variable.

(iii) A defining property is: "an object x is an element of the set iff it is one of the following integers: 2,3,5,7,11,13,17,19,23,29,31,37,41,43,47. This property can be expressed, as in Part (i), as $(x = 2) \vee (x = 3) \vee (x = 5) \vee \cdots \vee (x = 43) \vee (x = 47)$. Clearly, writing out a formula of such form is essentially the same as enumerating the elements in the set. When the number of elements in the set is huge, we will encounter the same problem as we have encountered in using the brace notation. A better approach is to define a predicate $prime$ such that:

$$prime(x) = \begin{cases} true & if\ x\ is\ a\ prime\ number \\ false & if\ x\ is\ not\ a\ prime\ number \end{cases}$$

An alternate defining property is thus: $prime(x) \wedge \sim (Greater(x, 50) \vee Equal(x, 50))$.

(iv) A defining property is: "an object x is an element of the set iff it is a province of Canada." Again, define a predicate $Canada$ such that:

$$Canada(x) = \begin{cases} true & if\ x\ is\ a\ province\ of\ Canada \\ false & if\ x\ is\ not\ a\ province\ of\ Canada \end{cases}$$

The defining property can then be expressed as $Canada(x)$.

(v) A defining property is: "an object x is an element of the set iff it is an university in North America." Let *university* be a predicate such that:

$$university(x) = \begin{cases} true & if\ x\ is\ an\ university\ in\ North\ America \\ false & if\ x\ is\ not\ an\ university\ in\ North\ America \end{cases}$$

The defining property can then be expressed as $university(x)$.

(vi) A defining property is: "an object x is an element of the set iff it is a rational number." Let *rational* be a predicate such that:

$$rational(x) = \begin{cases} true & if\ x\ is\ a\ rational\ number \\ false & if\ x\ is\ not\ a\ rational\ number \end{cases}$$

The defining property can then be expressed as $rational(x)$.

Using the fact that: "a number is a rational number iff it is a real number and is not an irrational number," we obtain yet another defining property for the set of rational numbers. This defining property can be expressed as $(x \in \mathbf{R}) \wedge \sim (irrational(x))$, where $\mathbf{R}$ is the set of real numbers, and *irrational* is a predicated such that

$$irrational(x) = \begin{cases} true & if\ x\ is\ an\ irrational\ number \\ false & if\ x\ is\ not\ an\ irrational\ number \end{cases}$$

Finally, a more elaborate defining property can be obtained by using the following property of rational numbers: "a number x is a rational number iff there exists two integers a and b such that $x = \frac{a}{b}$." Let $\mathbf{Z}$ be the set of all integers, we can express the defining property as $(\exists a, b)((a, b \in \mathbf{Z}) \wedge equal(x, a/b))$. Note that the second argument of the predicate *equal*, namely a/b, is an expression.

(vii) A defining property is: "an object p is an element of the set iff it is at a distance r from the fixed point o." Let $\mathbf{R}^2$ be the two-dimensional Euclidean space (i.e. the plane) and $d(x, y)$ denote the Euclidean distance between two points x and y on the plane. The above defining property can be expressed as $p \in \mathbf{R}^2 \wedge Equal(d(p, o), r)$. Here, the first argument of the predicate *Equal* is an expression. ■

Remarks

1. For the sake of clarity, in forming a formula, we should always have recourse to the most convenient form. It is for this reason that we often use a combination of formal and informal expressions in formulas. For example, we shall write "x is a rational number" as a formula and omit the explicit definition of the predicate *rational* completely.
2. Example 3.3.6(vi) shows that the defining property is non-unique. In general, any property that is logically equivalent to a defining property of a set can be used as a defining property of the same set.

Let $S(x)$ denote a formula containing a free variable x, and $S(a)$ denote the closed FOL wff resulting from $S(x)$ after every free occurrence of x is replaced by a a. We say that $S(x)$ is true for a if $S(a)$ evaluates to *true*.

Example 3.3.7 Let $S(x)$ be the formula $prime(x) \wedge \sim (Greater(x, 50) \vee Equal(x, 50))$ in Example 3.3.6(iii). Then, $S(7)$ is the closed FOL wff: $prime(7) \wedge \sim (Greater(7, 50) \vee Equal(7, 50))$. Since $prime(7) \wedge \sim (Greater(7, 50) \vee Equal(7, 50))$ is evaluated to the value *true*, the formula $prime(x) \wedge \sim (Greater(x, 50) \vee Equal(x, 50))$ is evaluated to true for the integer 7. i.e. $S(7)$ is evaluated to *true*.

Given a formula $S(x)$ in which x is the only free variable. If all those objects for which $S(x)$ evaluates to true form a set, then in the ***set-builder notation***, that set is denoted by

$$\{x \mid S(x)\}$$

read as "*the set of all x such that $S(x)$ is true.*"

Example 3.3.8 The following are specifications of the sets in Example 3.1.1 in set-builder notation, using the formulas given in Example 3.3.6. Note that we use informal expressions in some of the formulas.

(i) $\{x \mid (x = a) \vee (x = e) \vee (x = i) \vee (x = o) \vee (x = u)\}$

(ii) $\{v \mid (v = \{1, 2\}) \vee (v = \{a, b, c\}) \vee (v = \alpha) \vee (v = x) \vee (v = y) \vee (v = z)\}$

(iii) $\{x \mid prime(x) \wedge \sim (Greater(x, 50) \vee Equal(x, 50))\}$

(iv) $\{x \mid x\, is\, a\, province\, of\, Canada\}$

(v) $\{x \mid x\, is\, an\, university\, in\, North\, America\}$

(vi) $\{x \mid x\, is\, a\, rational\, number\}$, or $\{x \mid (\exists a, b)((a, b \in \mathbf{Z}) \wedge (x = a/b) \wedge (b \neq 0))\}$, where $\mathbf{Z}$ is the set of all integers, or $\{x \mid (x \in \mathbf{R}) \wedge \sim (irrational(x))\}$, where $\mathbf{R}$ is the set of all real numbers.

(vii) $\{p \mid p \in \mathbf{R}^2 \wedge d(p, o) = r\}$, where $\mathbf{R}^2$ is the two-dimensional Euclidean space (i.e. the plane) and $d(p,o)$ is the Euclidean distance between point p and the fixed point o.

Remarks

1. The FOL-wff $(\forall x)(x \in X \Rightarrow S(x))$ is commonly abbreviated as $(\forall x \in X)S(x)$, or $\forall x \in X, S(x)$, or $S(x), \forall x \in X$, or $S(x), for\, all\, x \in X$.

   The FOL-wff $(\exists x)(x \in X \wedge S(x))$ is commonly abbreviated as $(\exists x \in X)S(x)$, or $\exists x \in X, S(x)$, or $S(x), for\, some\, x \in X$.

   The following variant of FE7 and FE8 can be easily verified:

   FE7$_\in$: $\sim (\forall x \in X)S(x) \equiv (\exists x \in X) \sim S(x)$
   FE8$_\in$: $\sim (\exists x \in X)S(x) \equiv (\forall x \in X) \sim S(x)$

2. Since x is just a dummy variable in $S(x)$, for any variable y, let $S(y)$ be the formula resulting from $S(x)$ after every free occurrence of x in $S(x)$ is replaced by a y. Then $S(x) \equiv S(y)$ as long as $S(x)$ is free for y. As a result, $\{x \mid S(x)\}$ and $\{y \mid S(y)\}$ define the same set.

   For example, the property: "x is an even integer iff x=2y for some integer y" is a defining property of the set of all even integers. The latter can thus be defined as $\{x \mid (\exists y \in \mathbf{Z})(x = 2y)\}$. For the variable z, since there is no quantifier in the formula $(\exists y \in \mathbf{Z})(x = 2y)$ quantifying z, i.e. the formula $S(x)$ is free for z, the set of even integers can also be defined as $\{z \mid (\exists y \in \mathbf{Z})(z = 2y)\}$. However, the same set cannot be defined by $\{y \mid (\exists y \in \mathbf{Z})(y = 2y)\}$ because x does have an occurrence in $S(x)$ which is within the scope of a quantifier quantifying an occurrence of y. In other words, $S(x)$ is not free for y. As a matter of fact, the only integer y satisfying the equation y=2y is 0. As a result, $\{y \mid (\exists y \in \mathbf{Z})(y = 2y)\}$ is the singleton $\{0\}$.

3. The set-builder notation has a number of variants:

(i) Let $S(x)$ be a formula. Then $x \in A \wedge S(x)$ is also a formula. This is because $x \in A$ is a predicate as it can be rewritten as *memberof*(x,A) as follows:

$$memberof(x, A) = \begin{cases} true & \text{if } x \in A \\ false & \text{if } x \notin A \end{cases}$$

We will frequently encountered formulas of the form $x \in A \wedge S(x)$ where $S(x)$ is a formula. When this type of formulas is used in the set-builder notation, the resulting set $\{x \mid x \in A \wedge S(x)\}$ is customary written as $\{x \in A \mid S(x)\}$.

(ii) Let $f(y)$ be an expression containing y as a free variable. The set $\{x \mid (\exists y)(S(y) \wedge x = f(y))\}$ is customary written as $\{f(y) \mid S(y)\}$. For example, the set $\{x \mid (\exists y)(y \in \mathbf{Z} \wedge x = 2y)\}$ can be rewritten as $\{2y \mid y \in \mathbf{Z}\}$. More generally, let $f(y, z)$ be an expression containing y and z as free variables. The set $\{x \mid (\exists y)(\exists z)(S(y) \wedge S'(z) \wedge x = f(y, z))\}$ can be written as $\{f(y, z) \mid S(y) \wedge S'(z)\}$. An example is given in Example 3.3.9(iv) below.

The following are a few more examples of set specifications using the set-builder notation.

Example 3.3.9

(i) In set-builder notation, the singleton $\{a\}$ and the unordered pair $\{a, b\}$ can be specified as $\{x \mid x = a\}$ and $\{x \mid (x = a) \vee (x = b)\}$, respectively.

(ii) The set of positive real number less than 4 can be expressed as $\{x \mid x \in \mathbf{R} \wedge 0 \leq x < 4\}$ or $\{x \in \mathbf{R} \mid 0 \leq x < 4\}$, where $\mathbf{R}$ is the set of real numbers.

(iii) The set of all prime numbers can be specified as:

$$\{p \in \mathbf{Z}^+ \mid (p \geq 2) \wedge (\forall x \in \mathbf{Z}^+)((\exists y \in \mathbf{Z}^+)(p = x \cdot y) \Rightarrow (x = 1) \vee (x = p))\},$$

where $\mathbf{Z}^+$ is the set of all positive integers.

(iv) The set of the coordinates of all the points on the plane can be defined as $\{x \mid (\exists a)(\exists b)(a \in \mathbf{R} \wedge b \in \mathbf{R} \wedge x = (a, b))\}$, or $\{(a, b) \mid a \in \mathbf{R} \wedge b \in \mathbf{R}\}$, or $\{(a, b) \mid a, b \in \mathbf{R}\}$.

3.4 The Principle of Specification

A question naturally arises is: "Given any formula $S(x)$. Do all the objects x for which $S(x)$ is true always form a set?" In other words: "Given any formula $S(x)$. Does the set $\{x \mid S(x)\}$ always exist?" It turns out that the answer is *No*! The following is an example due to Russell.

The Russell's Paradox: $\sim (\exists X)(X = \{x \mid x \notin x\})$

i.e. Let $S(x)$ be '$x \notin x$.' Then the set $\{x \mid x \notin x\}$ does not exist.

(Proof by Contradiction) Suppose $(\exists X)(X = \{x \mid x \notin x\})$. Then $B = \{x \mid x \notin x\}$. (EI)

Now, we must have either $B \in B$ or $B \notin B$. We consider the two cases separately.

Case (i): Suppose $B \in B$. Then as $B = \{x \mid x \notin x\}$, B must satisfy the formula '$x \notin x$.' We thus have: $B \notin B$. But then $B \in B$ and $B \notin B \Rightarrow (B \in B) \wedge (B \notin B) \Rightarrow false$ (a contradiction).

Case (ii): Suppose $B \notin B$. Since $B = \{x \mid x \notin x\}$, the set B contains every x that satisfies the formula '$x \notin x$.' But $B \notin B$ implies that the set B satisfies the formula '$x \notin x$.' Therefore, $B \in B$. But then $B \in B$ and $B \notin B \Rightarrow (B \in B) \wedge (B \notin B) \Rightarrow false$ (a contradiction).

Since both cases lead to a contradiction, we thus conclude that $\sim (\exists X)(X = \{x \mid x \notin x\})$. In other words, the set $\{x \mid x \notin x\}$ does not exist. ■

Russell's paradox tells us that the set-builder notation, when used without restriction, can lead to contradictions. This puts us in a very awkward position, for given any formula, how can we tell if all the objects satisfying the formula do form a set. Fortunately, the *Principle of Specification* comes to our rescure.

The Principle of Specification: For every set A and every formula $S(x)$, there exists a set B whose elements are exactly those elements of A for which $S(x)$ is true. The set B is denoted by:

$$\{x \mid x \in A \wedge S(x)\} \text{ or } \{x \in A \mid S(x)\}.$$

The Principle of Specification tells us that while a mere formula may not determine a set, a formula with an existing set will always form a new set.

Intuitively, we can conceive of the existence of a set that contains *everything*. This set is commonly called the ***absolute universal set***. We shall prove that "*the absolute universal set does not exist.*" Formally, we shall prove that $\vdash \sim (\exists X)(\forall x)(x \in X)$

Proof: (Proof by Contradiction) Suppose $(\exists X)(\forall x)(x \in X)$.

Then $(\forall x)(x \in \mathrm{U})$, where U is a set (a constant). (EI) . . . (I)

By the Principle of Specification, $(\exists Y)(Y = \{x \in \mathrm{U} \mid x \notin x\})$.

$\Rightarrow B = \{x \in \mathrm{U} \mid x \notin x\}$ (EI), B is a set (a constant).

It follows that $B \in \mathrm{U}$. ((I),UI) . . . (II)

Now, we have either $B \in B$ or $B \notin B$.

(i) Suppose $B \in B$ (i.e. B is an element of B). . . . (III)

Then $B \in \mathrm{U} \wedge B \notin B$ $(\because B = \{x \in \mathrm{U} \mid x \notin x\})$

$\Rightarrow B \notin B \wedge B \in \mathrm{U}$ (E9)

$\Rightarrow B \notin B$ (I2) . . . (IV)

$\Rightarrow B \in B \wedge B \notin B$ ((III),(IV),I6)

$\Rightarrow false$ (E1)

(ii) Suppose $B \notin B$. . . . (V)

Then $B \in \mathrm{U} \wedge B \notin B$ ((II),(V),I6)

$\Rightarrow B \in B$ $(\because B = \{x \in \mathrm{U} \mid x \notin x\})$ . . . (VI)

$\Rightarrow B \in B \wedge B \notin B$ ((VI),(V),I6)

$\Rightarrow false$ (E1)

Since in either case, we have a contradiction, we therefore conclude that $\vdash \sim (\exists X)(\forall x)(x \in X)$. i.e., the absolute universe set does not exist. ■

3.5 The Empty Set

Definition: The set $\{x \mid (x = x) \wedge (x \neq x)\}$ is called the ***empty set*** and is denoted by $\emptyset$.

Since the formula $(x = x) \wedge (x \neq x)$ is always false, nothing satisfies the formula. Therefore, the empty set contains *no* element. This is stated more formally in the following lemma.

Lemma 3.5.7: $x \in \emptyset \Leftrightarrow false$.

Proof: (Bidirectional Proof)

$x \in \emptyset \Leftrightarrow (x = x) \wedge (x \neq x)$ (Definition of $\emptyset$)
$\Leftrightarrow false$ (E1) ■

Lemma 3.5.8: $A \neq \emptyset \Leftrightarrow (\exists x) x \in A$.

Proof: $A \neq \emptyset \Leftrightarrow (\exists x)(x \in A \wedge x \notin \emptyset) \vee (\exists x)(x \notin A \wedge x \in \emptyset)$ (Lemma 3.2.2)
$\Leftrightarrow (\exists x)((x \in A \wedge \sim x \in \emptyset) \vee (x \notin A \wedge x \in \emptyset))$ (FE10)
$\Leftrightarrow (\exists x)((x \in A \wedge \sim false) \vee (x \notin A \wedge false))$ (Lemma 3.5.7)
$\Leftrightarrow (\exists x)((x \in A \wedge true) \vee (x \notin A \wedge false))$ (Theorem 1.7.2(i))
$\Leftrightarrow (\exists x)(x \in A \vee (x \notin A \wedge false))$ (E5)
$\Leftrightarrow (\exists x)(x \in A \vee false)$ (E7)
$\Leftrightarrow (\exists x) x \in A$ (E6) ■

Corollary 3.5.8.1 $\sim (\exists x) x \in \emptyset$.

Proof: (Proof by contradiction)
Suppose $(\exists x) x \in \emptyset$.
Then $a \in \emptyset$ (EI) $\cdots\cdots$ (I)
By Lemma 3.5.7, $(\forall x)(x \in \emptyset \Leftrightarrow false)$ (Recall that the free variable x is quantified by $\forall$)
$\Rightarrow a \in \emptyset \Leftrightarrow false$ (UI)
$\Rightarrow (a \in \emptyset \Rightarrow false) \wedge (false \Rightarrow a \in \emptyset)$ (E20)
$\Rightarrow (a \in \emptyset \Rightarrow false)$ (I2) $\cdots\cdots$ (II)
$\Rightarrow false$ ((I),(II),I3) ■

The defining property, $(x = x) \wedge (x \neq x)$, used in the definition of $\emptyset$ is equivalent to the logical constant *false*. Therefore, any formula that is equivalent to *false* can be used as a defining property for $\emptyset$. For instance, as $(x \in x) \wedge (x \notin x) \Leftrightarrow false$, the empty set $\emptyset$ could have been defined as $\{x \mid (x \in x) \wedge (x \notin x)\}$.

Theorem 3.5.9: $\emptyset \subseteq A$, *for any set* A.

Proof: We shall prove that $(\forall x)(x \in \emptyset \Rightarrow x \in A)$.
(Vacuous Proof) $x \in \emptyset \Leftrightarrow false$ (Lemma 3.5.7)
$\Rightarrow (x \in \emptyset \Rightarrow false) \wedge (false \Rightarrow x \in \emptyset)$ (E20)
$\Rightarrow (x \in \emptyset \Rightarrow false)$ (I2)
Therefore, $(x \in \emptyset \Rightarrow x \in A)$.
It follows that $(\forall x)(x \in \emptyset \Rightarrow x \in A)$ (Gen)
$\Rightarrow \emptyset \subseteq A$ (Definition of $\subseteq$) ■

3.6 Power Sets

Definition: Let A be a set. The ***power set*** of A, denoted by $\mathcal{P}(A)$, is the set $\{X \mid X \subseteq A\}$.

Therefore, the power set of a given set is the set consisting of all the subsets of the given set.

Example 3.6.10

(i) Let $A = \{a\}$. Then $\mathcal{P}(A) = \{\emptyset, \{a\}\}$.

(ii) Let $A = \{a, b\}$. Then $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.

(iii) Let $A = \emptyset$. Then $\mathcal{P}(A) = \{\emptyset\}$.

(iv) Let $A = \{1, \{1, 2\}, \{3\}\}$.

Then $\mathcal{P}(A) = \{\emptyset, \{1\}, \{\{1, 2\}\}, \{\{3\}\}, \{1, \{1, 2\}\}, \{1, \{3\}\}, \{\{1, 2\}, \{3\}\}, \{1, \{1, 2\}, \{3\}\}\}$.

(v) Let A be a set of n elements. Then $\mathcal{P}(A)$ contains 2^n elements.

Proof: (By Mathematical Induction on n). ■

A power set is an example of a set of sets. We shall called a set of sets a ***collection of sets***.

3.7 Examples

1. Prove that: if $A \subseteq B$ and $B \subset C$, then $A \subset C$.

Proof: (Direct Proof) Suppose $A \subseteq B$ and $B \subset C$. . . . (I)

We are to prove that $A \subset C$. By definition of $\subset$, this is equivalent to proving that $A \subseteq C$ and $A \neq C$.

First, we shall prove that $A \subseteq C$.

$B \subset C$ (by (I))

$\Rightarrow (B \subseteq C) \wedge (B \neq C)$ (Definition of $\subset$)

$\Rightarrow (B \subseteq C)$ (I2) . . . (II)

$\Rightarrow (A \subseteq B) \wedge (B \subseteq C)$ ((I),(II),I6)

$\Rightarrow A \subseteq C.$ (Theorem 3.2.3(iii)) . . . (A)

Next, we shall prove that $A \neq C$.

$B \subset C$ (by (I))

$\Rightarrow (\exists y)(y \in C \wedge y \notin B)$ (Lemma 3.2.6)

$\Rightarrow a \in C \wedge a \notin B$ (EI), a is a constant

$\Rightarrow a \in C$ and $a \notin B$ (E9, I2) . . . . . (III)

$A \subseteq B$ (by (I))

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in B)$ (Definition of $\subseteq$)

$\Rightarrow (a \in A \Rightarrow a \in B)$ (UI) . . . . . (IV)

$\Rightarrow a \notin A$ ((III),(IV),I4) . . . (V)

$\Rightarrow a \notin A \wedge a \in C$ ((V),(III),,I6)

$\Rightarrow (\exists x)(x \notin A \wedge x \in C)$ (EQ)

$\Rightarrow (\exists x)(x \notin A \wedge x \in C) \vee (\exists x)(x \in A \wedge x \notin C)$ (I1)

$\Rightarrow (\exists x)(x \in A \wedge x \notin C) \vee (\exists x)(x \notin A \wedge x \in C)$ (E10)

$\Rightarrow A \neq C$ (Lemma 3.2.2) . . . (B)

Hence, $(A \subseteq C) \wedge (A \neq C)$ ((A),(B),I6)

$\Rightarrow A \subset C$ (Definition of $\subset$) ■

3.8 Exercises

1. Prove that:
   (a) if $A \subset B$ and $B \subset C$, then $A \subset C$.
   (b) if $A \subset B$ and $B \subseteq C$, then $A \subset C$.
2. Present a set which contains at least three elements and in which every element is also a subset of the set.
3. Prove or disprove the following: if $A \not\subseteq B$ and $B \not\subseteq C$, then $A \not\subseteq C$.
4. Prove that the set of all sets does not exist.
5. Prove that if $A \subseteq B$ then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.
6. What are the elements of the power set of $\{x, \{y, z\}\}$? How many elements does this power set have?
7. Let $B \subseteq A$ and $\mathcal{P}(A : B) = \{X \in \mathcal{P}(A) \mid B \subseteq X\}$.
   (a) Let $A = \{a, b, c, d, e\}$ and $B = \{a, b\}$. List all the elements of the set $\mathcal{P}(A : B)$. How many are there?
   (b) Prove that $\mathcal{P}(A : \emptyset) = \mathcal{P}(A)$.
8. Let $A_i, 1 \leq i \leq n$, be sets. Prove that: $A_i \subseteq A_{i+1}, 1 \leq i < n$, and $A_n \subseteq A_1$ iff $(\forall i)(\forall j)(1 \leq i < j \leq n \Rightarrow A_i = A_j)$.
9. Prove that:
   (a) $\sim (\forall x \in X)S(x) \equiv (\exists x \in X) \sim S(x)$
   (b) $\sim (\exists x \in X)S(x) \equiv (\forall x \in X) \sim S(x)$